MILITARY SPECIFICATION

TIE-DOWN ASSEMBLY, CARGO (MUST)

1. SCOPE

1.1 This specification covers a cargo tie-down assembly consisting of four tie-down strap assemblies, one nylon web net assembly and one nylon rope net assembly (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

T-T-871	-	Twine, Cotton, Wrapping
V-T-295	-	Thread, Nylon
PPP-B-636	-	Boxes, Shipping, Fiberboard

MILITARY

MIL-R-17343	-	Rope, Nylon
MIL-W-27265	-	Webbing, Textile, Woven Nylon, Impregnated

STANDARDS

FEDERAL

FED-STD-751	-	Stitches, Seams, and Stitching

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-130	-	Identification Marking of U.S. Military Property

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DRAWINGS

ARMY NATICK LABORATORIES

5-4-1661 - Tie-Down Assembly, Cargo (MUST); Tie-Down Strap Assembly
5-4-1662 - Web Net Assembly
5-4-1663 - Rope Net Assembly

(Miniature copies of Drawings 5-4-1661, 5-4-1662 and 5-4-1663, identified as figures 1, 2, and 3, are attached for information only.)

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Tariff Order Section, 1616 P Street, N.W., Washington, D.C. 20036.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the supplier shall furnish a sample of the item to be procured under this specification for first article inspection and approval (see 4.3 and 6.3).

3.2 Materials and components. Materials and components shall be in accordance with the specifications, standards and drawings listed in section 2 and as specified herein.

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3.2.1 Alternative components. Components offered as equivalent to components specified hereinafter and on the referenced drawings as a specific manufacturer's part number or equal shall be functionally equal of equal or better quality, with the part number identified. The incorporation and inclusion of such a component in the design of the specified end product shall not require modification or change to any other specified component, and not reduce ease of maintenance to it or to any other components, unless such modification or change is specifically approved by the contracting officer. Prior to manufacture of the preproduction first article, or if none is required, prior to commencing production, the contractor shall submit for the contracting officer's approval a list identifying each proposed alternate component, together with proof that each listed component complies with requirements specified herein. The contracting officer, at his option, may require a physical sample of the proposed substitution. Approval of the submitted listing, together with supporting data, authorizes the commencement of the first article or production manufacture, as applicable, and does not relieve the supplier of the responsibility that these components perform in accordance with specified requirements when incorporated into the end product.

3.3 Construction. Tie-down assemblies shall conform to the drawings listed in section 2 and as specified herein.

3.3.1 Stitching, machine. All stitching shall conform to type 301 of FED-STD-751, 4 to 6 stitches per inch.

3.3.1.1 Thread breaks. Thread breaks in stitching shall be overstitched not less than 1 inch at each break.

3.3.1.2 Skipped stitches. Two or more consecutively skipped stitches shall be overstitched not less than 1 inch.

3.3.1.3 Runoffs. Runoffs shall be restitched to conform to the required stitch pattern.

3.3.1.4 Stitching ends. Ends of stitching shall be overstitched not less than 1 inch except where ends are turned under in a hem, held down by other stitching, or where stitching is performed automatically. Where stitching is performed automatically on stitch patterns such as box or box with cross-stitch, three or more tying, overlapping or backstitches shall be used to secure the ends.

3.3.1.5 Fusing of webbing ends. Cut ends of webbing shall be fused and, after fusing, shall be free from frayed edges and loose fibers.

3.4 Identification marking. Identification marking on each tie-down assembly component shall conform to MIL-STD-130.

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3.5 Workmanship. Webbing shall contain no frayed or scalloped edges. Rope components shall be free of cuts or abrasions. Thread tension shall be maintained so that there shall be no loose, or tight, stitching. Stitching margins shall be maintained so that there shall be no runoffs. Care shall be taken in sewing to insure that there shall be no needle chews. All thread ends shall be trimmed to 1/2 inch or less. Metal components shall be free of burrs, sharp edges or corroded areas.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The supplier shall insure that all components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards unless excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).
3. Inspection of preparation for delivery (see 4.5).

4.3 First article inspection. When a first article is required, the tie-down assembly shall be examined for defects listed in table I and dimensions specified. The presence of any defect or any dimension not within the specification tolerance shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105.

4.4.1 End item inspection. The lot shall consist of all tie-down assemblies offered for inspection at one time. The sample unit shall be one tie-down assembly.

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4.4.1.1 Visual examination. The tie-down assembly shall be examined for the defects in table I. The inspection level shall be II with acceptable quality levels (AQL's), expressed in terms of defects per hundred units, of 1.5 major and 10.0 total, major and minor defects combined.

TABLE I. Classification of defects for visual characteristics

Examine	Defects	Classification Major	Minor
Webbing	Edge frayed or scalloped	X	
	Cut end not fused	X	
	Cut end fused but contains frayed edge or loose fibers		X
Rope components	Cut	X	
	Abrasion		X
Hardware	Broken	X	
	Malformed:		
	- fails to perform intended function	X	
	- but will perform intended function		X
	Corroded area	X	
	Burr or sharp edge which may cause injury in handling or damage to webbing	X	
Stitching	Wrong stitch type	X	
	Any open seam	X	
NOTE:	A seam shall be classified as open when one or more stitches joining a seam are broken or when two or more skipped or runoff stitches occur. Repairs of open seams shall not be scored as defects.		
	Thread breaks overstitched less than 1 inch		X
	Needle chews resulting in cut or hole	X	
	Skipped stitches overstitched less than 1 inch (where 2 or more consecutively skipped stitches occur)		X
NOTE:	Skipped stitches not overstitched shall be classified as open seams.		

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TABLE I. Classification of defects for visual characteristics (cont'd)

Examine	Defects	Classification Major	Minor
Stitch tension	Loose, resulting in an exposed bobbin or top thread; or, tight resulting in breaking of thread when normal tension is applied.	X	
Stitches per inch	Less than minimum specified:-		
	- one stitch		X
	- two or more stitches	X	
	More than maximum specified:		
	- one stitch		X
	- two or more stitches	X	
Stitching ends	Overstitched less than 1 inch (except where ends are held down by other stitching or turned under in a hem or where stitching is performed automatically).		X
	Secured with less than three tying, overlapping or backstitches when automatic stitching is performed.		X
Components and assembly	Any component part not fabricated of applicable referenced material	X	
Cleanness	Grease or oil stain; thread ends not trimmed to 1/2 inch or less throughout.		X
Marking	Missing, incorrect, illegible, misplaced or size characters not as specified.		X

4.4.1.2 Dimensional examination. The tie-down assemblies shall be examined for compliance with dimensions specified. Any dimension not within the specification tolerance shall be a defect. The inspection level shall be S-3 with an AQL of 4.0 expressed in terms of defects per hundred units.

4.5 Examination of preparation for delivery requirements. An examination shall be made to determine that packaging and packing and marking are in compliance with the requirements of section 5 of this specification. Defects shall be scored in accordance with table II. The sample unit shall be one

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shipping container fully prepared for delivery. The lot shall be the number of shipping containers offered for inspection at one time. The inspection level shall be S-2 and the AQL shall be 2.5, expressed in terms of defects per hundred units.

TABLE II. Examination of preparation for delivery

Examine	Defects
Marking	Omitted, incorrect, illegible, of improper size, location, sequence or method of application
Materials	Any component missing or damaged
Workmanship	Bulging or distortion of containers

4.5.1 Examination of shipping containers. When shipping containers are required to be in accordance with PPP-B-636, examination for defects in closure, waterproofing and reinforcing shall be in accordance with that specification.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Tie-down strap assemblies. The six straps of each tie-down strap assembly shall be neatly and compactly coiled onto their respective tie-down buckle and secured with twine conforming to type I or II, 8-ply of T-T-871. Four tie-down strap assemblies shall then be packaged in a snug-fitting fiberboard box conforming to type CF (variety SW) or SF, class domestic, grade 200 of PPP-B-636. Each box shall be closed in accordance with the appendix of PPP-B-636.

5.1.1.2 Net assemblies. One web net and one rope net shall be packaged in a snug-fitting fiberboard box conforming to type CF (variety SW) or SF, class domestic, grade 200 of PPP-B-636. Each box shall be closed in accordance with the appendix of PPP-B-636.

5.1.2 Level C. Cargo tie-down assemblies shall be packaged to afford adequate protection against physical damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

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5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Each complete cargo tie-down assembly, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to grade V2s, style RSC of PPP-B-636. Closure, waterproofing by means of tape, and reinforcing shall be in accordance with the appendix of PPP-B-636.

5.2.2 Level B. Each complete cargo tie-down assembly, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to type CF (variety SW), or SF, class domestic, style RSC of PPP-B-636. Closure shall be in accordance with method II of the appendix of PPP-B-636.

5.2.2.1 When specified (see 6.2), the shipping container shall be a grade V3c, V3s or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix thereto.

5.2.3 Level C. Cargo tie-down assemblies packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules as applicable.

5.3 Marking. In addition to any special marking required by the contract or order, interior boxes and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The cargo tie-down assembly covered by this specification is intended for use in the Ward Container-Patient Service Element and the Expandable Shelter of the Medical Unit, Self Contained Transportable (MUST) system for cargo restraint during shipment.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. When a first article is required (see 3.1, 4.3 and 6.3).
c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
d. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).

6.3 First article. When a first article is required it shall be inspected and approved under appropriate provisions of ASPR 7-104.55. The first article should be a preproduction sample, initial production item or other specific item described under definition of first article in ASPR. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

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6.4 Supersession data. This specification includes the requirements of Army Natick Laboratories Limited Production Purchase Description LP/P. DES 53-69.

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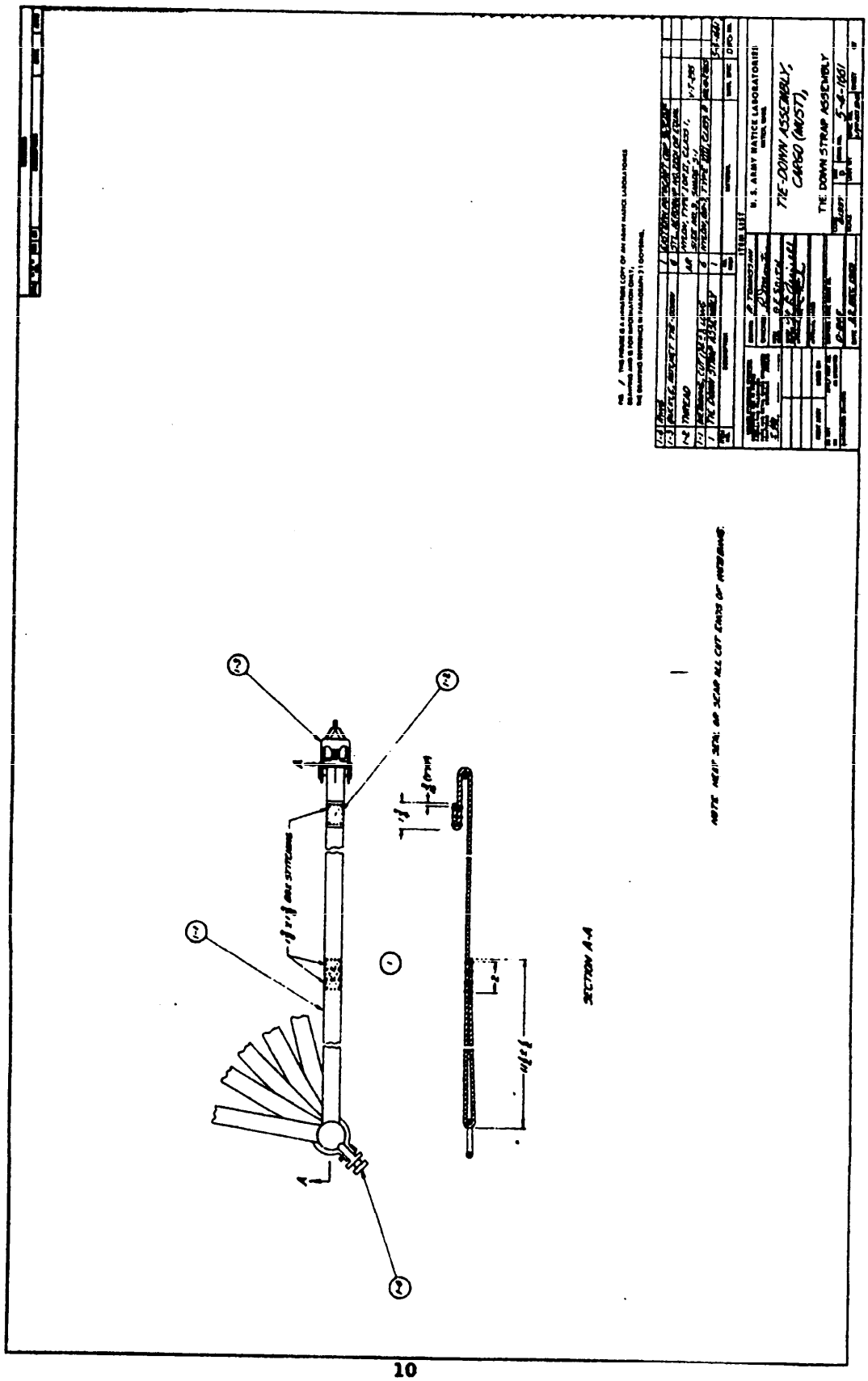
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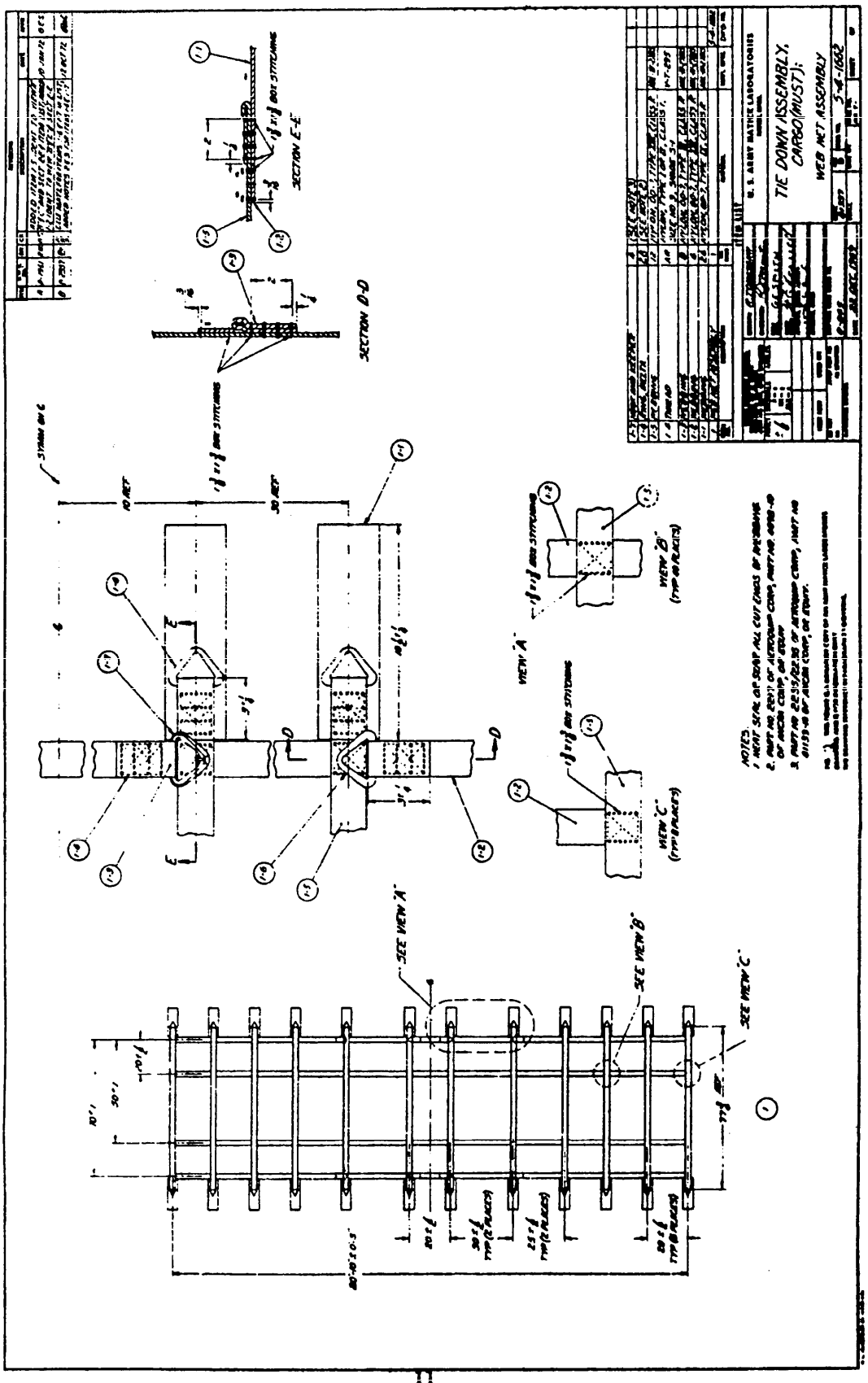
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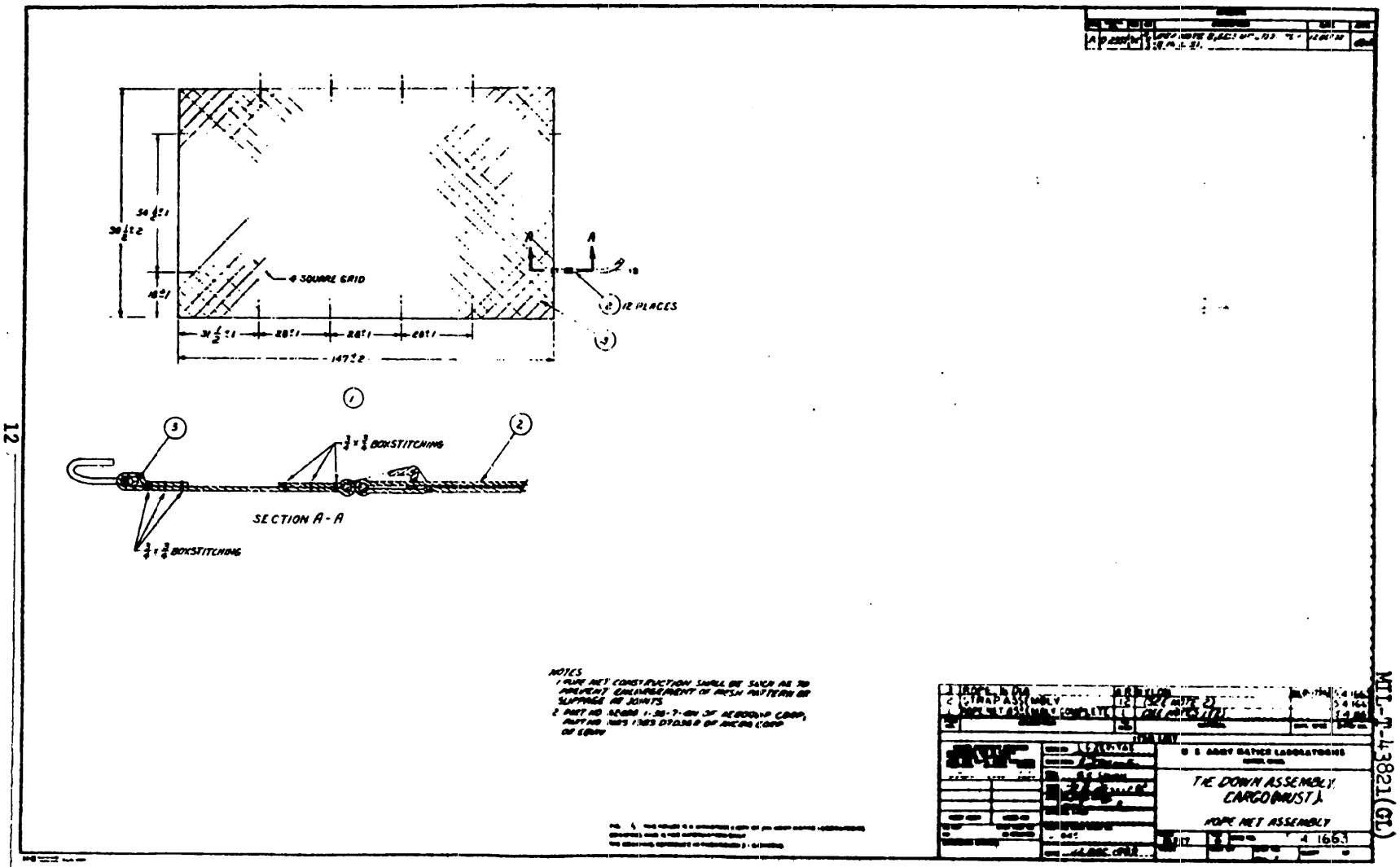
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